Clinical Document Intelligence Hub

Objective

Design and develop an AI prototype that can ingest unstructured clinical documents and surface structured, actionable intelligence. The solution should demonstrate how AI can transform fragmented healthcare data into consistent, decision-ready outputs — reducing the manual burden on clinical and administrative teams.

Context

You are working with a mid-size healthcare provider whose clinical and administrative staff spend significant time manually reviewing patient documents — intake forms, discharge summaries, lab reports, and physician notes — to extract key information. This process is slow, inconsistent, and prone to error. The client wants a proof-of-concept that shows how AI can read these documents, extract structured data, and generate meaningful summaries or recommendations.

Your goal is to design and demonstrate a working prototype that addresses this challenge within 5 days.

Requirements and Deliverables

Deliverable	Description
Core Functionality	Build a working prototype that accepts clinical document inputs (text, PDF, or image), extracts key structured information, and generates a clear, readable summary or recommendation relevant to the healthcare context.
Optional Enhancement	Candidates may explore confidence scoring on extracted fields, multi-document comparison, or integration with a simple triage or routing logic to demonstrate additional value.
Output Format	The prototype should produce a structured output — such as a patient summary card, a risk flag, or a recommended next step — presented clearly for a clinical or administrative audience.
Documentation	A one-page README describing the approach, the AI models or tools used, assumptions made, and one example of input and output.

Candidates are free to use any publicly available dataset or generate a suitable dataset using AI tools. No proprietary or client data is required.

Suggested Tools and Platforms

Pro-Code Track

- Python with an LLM API (OpenAI, Claude, or similar) for document parsing and summarization
- Streamlit or Gradio for a simple UI; PDF/image parsing libraries as needed

No-Code Track

- Lyzr Studio, Retool, or n8n with an AI vision or document node
- Optional: Google Drive or Dropbox for document input

Feel free to use the tools you enjoy working with. We do not cover subscription or license costs — please explore free trials or open-source options.

Submission Format

Each candidate must submit three components within 5 days of receiving the brief.

1. Working Demo

- Demonstrate the prototype end-to-end: document input → AI processing → structured output
- Accepted formats: hosted application link, or screen recording (maximum 3 minutes)
- The demo should show a real or realistic document input, the generated output, and a brief explanation of the approach

2. Five-Slide Summary Deck (Fixed Structure)

Slide	Title	Description
1	PROBLEM UNDERSTANDING AND OBJECTIVE	Summarize your understanding of the brief and the problem statement.
2	SOLUTION ARCHITECTURE AND DESIGN FLOW	Present your workflow or system design. Include a diagram showing major components and data flow.
3	IMPLEMENTATION HIGHLIGHTS	Highlight technical decisions, AI logic, or workflow details. Include screenshots or concise code snippets.
4	CHALLENGES AND LEARNINGS	Discuss key technical or design challenges, trade-offs, and key takeaways from the build.
5	DEMO SUMMARY AND NEXT STEPS	Present your final solution, include demo or repository links, and describe potential enhancements if given more time.

Format: PowerPoint or PDF (five slides maximum).

3. Supporting Assets

- README file with setup instructions and design notes
- Sample input and output (screenshots or text)
- Optional: GitHub repository or workspace link

Evaluation Rubric

Evaluation Focus	Weight
AI integration quality; accuracy and usefulness of extracted or generated output	40%

Relevance and clarity of the structured output; quality of clinical reasoning demonstrated	30%
Communication clarity and presentation structure	15%
Creativity, initiative, and reflection shown in documentation	15%

Deadline and Submission

- Time Limit: 5 days from the time this brief is issued
- Submission Package: demo link or short video recording, five-slide summary deck (PDF or PPT), README or supporting documentation
- Submissions should be delivered to the specified email address before midnight of the 5th day